Climatesimulation ‘game’



goal

- teach people how simulations/ models work
- teach people the global atmospheric circulation, global energy exchange and greenhouse effect in a very simplified manner

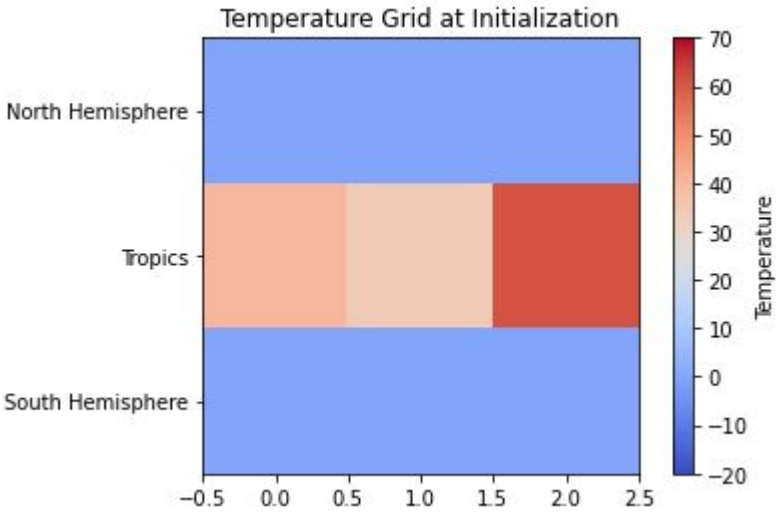
‘game’ components

- 9 players and a game master
- each player has piece of a paper and optionally a calculator



‘game’ setup

- players stand in a 3x3 formation
  - the two outer rows are the north/ south hemisphere
  - the middle row represents the equator/ tropics
- the hemispheres start with 0 temperature ‘points’
- the first two tropics positions choose a temperature between 25 and 40; the third tropics position get 135-temp_1-temp_2
  - here: [40., 34., 61.]



‘game’ rules and play

1. wind simulation:
    a. the hemispheres move one step to the right (jet streams)
    b. the tropics move one step to the left (trade winds)
2. temperature/ energy exchange between tropics and hemispheres:
    a. each tropics position chooses the hemisphere to which the difference is larger. If the difference is equal a random decision can be made
    b. it then can choose how much temperature it wants to exchange: a random number between 0 and half of the difference
3. temperature/ energy exchange between earth and space:
    a. each tropics position gets 8 temperature points from the sun
    b. each hemisphere position loses 4 temperature points to space
4. the gamemaster writes down the temperature points of each position and calculates the mean

gameplay twist

- global warming:
  a. at some point the temperature/ energy exchange between earth and space changes. Now the hemispheres loose 3 temperature points instead of 4 in every iteration.

discussion

- does it make sense?
- is it understandable enough?
- what could be added/ altered (without further complicating it)?